

25-Unity-AI-Vis

Author: Andreas Bueckle

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Use <https://kepler.gl/demo/nyctrips?>

Code:

I built this **Historical Reconstruction in VR** Project where users can explore a virtual model of the Sphinx of Giza and **interact with an AI-powered NPC tour guide**. Since there wasn't a plug-and-play solution for the NPC, I built the entire system from scratch using:

- DAZ 3D Studio for creating and morphing the character model
- Blender for 3D modeling
- Unity for the VR development and integration
- OpenAI Whisper Large V3 for Speech-to-Text
- LLaMa 3 (8b, 8192) via GroqAI for the LLM
- Simba via Speechify for Text-to-Speech
- And Unity's BlendShapes and Audio Pipeline for lip-sync and real-time responses.

Here's the final video demo of the project, would love to hear your thoughts and feedback on it!

 [Historical Reconstruction VR - Made with Clipchamp.mp4](#)

 [Historical Reconstruction VR - Made with Clipchamp.mp4](#)

Thanks a lot, and looking forward to catching up soon!

Research Plan

Introduction: Recent advances in immersive analytics¹⁻⁸ have demonstrated VR's potential for exploring complex spatial datasets across disciplines—from networks⁹ to mining shafts¹⁰ to protein structures¹¹. However, current VR data exploration suffers from significant cognitive overhead as users must mentally parse 3D visualizations while simultaneously formulating analytical questions. Traditional AI assistants remain confined to 2D interfaces, creating a disconnect between spatial data exploration and intelligent query support. This gap is particularly problematic for high-dimensional datasets where pattern recognition requires both spatial reasoning and analytical guidance. While embodied conversational agents exist in gaming contexts (usually as non-playable characters), their application to data visualization remains unexplored, despite evidence demonstrating enhanced retention of information in distinctive virtual contexts¹². Moreover, the field lacks empirical understanding of how different AI embodiments affect user performance in immersive analytical tasks, which represents a critical research opportunity at the intersection of data visualization, HCI, and data science.

Specific Aims: This project develops **VR-AI-ment**, a pilot application to jumpstart further research into how AI embodiment affects analytical performance in immersive data environments. The application will feature two AI embodiments supporting the user while they are analyzing building sensor data: a **humanoid character (E1)** and an **animated particle swarm (E2)**. VR-AI-ment will enable users to perform structured analytical tasks while task completion time, query accuracy, and cognitive load can be measured through established HCI metrics. This work addresses a fundamental gap in understanding optimal AI presentation for immersive analytics, providing evidence-based guidelines for future applications. While no user study is planned for this award mechanism, the VR-AI-ment application will be utilized as a proof of concept to acquire larger, external funding for a subsequent formal user study.

Preliminary Data: This work builds on PI Bueckle's established VR research program, including published user studies on optimizing manipulation¹³ and navigation¹⁴ as well as biomedical visualization¹⁵ in VR environments. The Data Visualization Literacy Framework (DVL-FW)¹⁶, co-authored by PI Bueckle, provides validated theoretical foundations for designing effective spatial visualizations. A functional prototype repository for VR-AI-ment exists on GitHub (<https://github.com/cns-iu/cns-unity-ai-vr>), demonstrating basic VR-AI integration. The prototype was implemented with [Unity 3D](#), a game engine to create real-time 3D applications for VR headsets. **Fig. 1** shows preliminary visualizations of both embodiments overlaid on sensor data from a 3D building model to illustrate the spatial layout of VR-AI-ment. Key Personnel Vo will contribute validated spatial analysis expertise and access to real building sensor datasets to ensure ecological validity of our test scenarios.

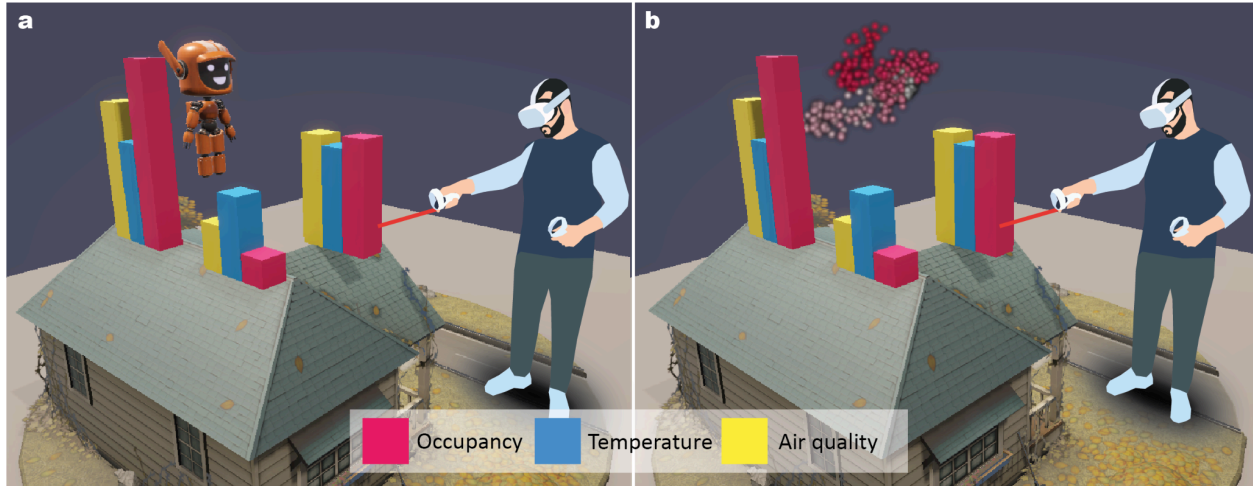


Fig. 1. Preliminary visualization of a 3D scene involving a user in VR-AI-ment. **a:** The user inspects a 3D bar graph visualization on top of a 3D model of a house with a humanoid AI embodiment. **b:** This scene uses the animated particle swarm. For this pre-visualization, freely available test assets were used from Sketchfab, an online 3D model repository; concretely, a [house](#) and a [robot](#) were used.

Design and Methods: The project will achieve its aims through systematic development and continuous informal user evaluation over 9 months:

Main Aim - Develop VR-AI-ment: The application will present users with a 3D building model overlaid with sensor data (occupancy, temperature, occupancy) provided by Key Personnel Vo. Users will be able to navigate the immersive environment using Meta Quest VR headsets while querying an AI assistant about patterns, correlations, and anomalies. The system integrates real-time speech recognition, natural language processing, and spatial data analysis to provide contextual responses. Both embodiments will answer identical queries to enable controlled comparison. **Fig. 1** demonstrates the core interaction paradigm, where users explore 3D visualizations while conversing with spatially-embedded AI assistants about data patterns and relationships.

Sub-Aim 1 - Humanoid Embodiment (E1): A carefully selected robot character will provide familiar anthropomorphic interaction cues through gestures, gaze direction, and lip synchronization. The model will be optimized for VR performance while maintaining social presence indicators that enhance trust and engagement in human-AI interaction.

Sub-Aim 2 - Animated Particle Swarm Embodiment (E2): Unity 3D's particle system, typically used to make fire, smoke, and rain effects, will generate responsive visual clusters that move, pulse, and reorganize based on AI responses. This abstract embodiment eliminates anthropomorphic bias while maintaining spatial presence and visual feedback to test whether functional embodiment can achieve similar user outcomes without social cognitive overhead.

Implementation: Technical development will use Unity 3D for VR integration and Meta Quest compatibility, with [Blender](#) for 3D modeling, integrating OpenAI's [Whisper](#) for speech recognition, [Llama 3](#) for natural language processing, and [Speechify](#) for response synthesis. Real-time performance optimization will ensure <100ms response latency to maintain immersive flow. An experienced Hourly Student Research Assistant will be hired from SICE's robust VR curriculum, which includes classes with a focus on virtual environments and 3D modeling, such as INFO-I 304 Introduction to Virtual Reality, INFO-I

References

1. Batch, A. *et al.* There Is No Spoon: Evaluating Performance, Space Use, and Presence with Expert Domain Users in Immersive Analytics. *IEEE Trans. Vis. Comput. Graph.* **26**, 536–546 (2020).
2. Billinghamurst, M., Cordeil, M., Bezerianos, A. & Margolis, T. Collaborative Immersive Analytics. in *Immersive Analytics* (eds Marriott, K. *et al.*) vol. 11190 221–257 (Springer International Publishing, Cham, 2018).
3. Chandler, T. *et al.* Immersive Analytics. in *2015 Big Data Visual Analytics (BDVA)* 1–8 (IEEE, Hobart, Australia, 2015). doi:10.1109/BDVA.2015.7314296.
4. Dwyer, T. *et al.* Immersive Analytics: An Introduction. in *Immersive Analytics* (eds Marriott, K. *et al.*) 1–23 (Springer International Publishing, Cham, 2018). doi:10.1007/978-3-030-01388-2_1.
5. Cordeil, M. *et al.* IATK: An immersive analytics toolkit. in 200–209 (IEEE, Washington, DC, 2019). doi:10.1109/VR.2019.8797978.
6. Filho, J. A. W., Freitas, C. M. D. S. & Nedel, L. Comfortable Immersive Analytics With the VirtualDesk Metaphor. *IEEE Comput. Graph. Appl.* **39**, 41–53 (2019).
7. *State-of-the-Art Survey: Immersive Analytics*. vol. 11190 (Springer International Publishing, Cham, 2018).
8. Cordeil, M., Cunningham, A., Dwyer, T., Thomas, B. H. & Marriott, K. ImAxes: Immersive Axes as Embodied Affordances for Interactive Multivariate Data Visualisation. in *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology* 71–83 (Association for Computing Machinery, New York, NY, USA, 2017). doi:10.1145/3126594.3126613.

9. Pirch, S. *et al.* The VRNetzer platform enables interactive network analysis in Virtual Reality. *Nat. Commun.* **12**, 2432 (2021).
10. Kaiser, P., Vasak, P., Suorineni, F. & Thibodeau, D. New Dimensions in Seismic Data Interpretation with 3-D Virtual Reality Visualisation for Burst-Prone Mines. in (eds Potvin, Y. & Hudyma, M.) 33–45 (Australian Centre for Geomechanics, Perth, 2005).
11. Cassidy, K. C., Šefčík, J., Raghav, Y., Chang, A. & Durrant, J. D. ProteinVR: Web-based molecular visualization in virtual reality. *PLOS Comput. Biol.* **16**, e1007747 (2020).
12. Parong, J. & Mayer, R. E. Cognitive and affective processes for learning science in immersive virtual reality. *J. Comput. Assist. Learn.* **37**, 226–241 (2021).
13. Bueckle, A., Buehling, K., Shih, P. C. & Börner, K. 3D virtual reality vs. 2D desktop registration user interface comparison. *PLOS ONE* **16**, e0258103 (2021).
14. Bueckle, A., Buehling, K., Shih, P. C. & Börner, K. Optimizing Performance and Satisfaction in Matching and Movement Tasks in Virtual Reality with Interventions Using the Data Visualization Literacy Framework. *Front. Virtual Real.* **2**, (2022).
15. Bueckle, A. *et al.* The HRA Organ Gallery affords immersive superpowers for building and exploring the Human Reference Atlas with virtual reality. *Front. Bioinforma.* **3**, (2023).
16. Börner, K., Bueckle, A. & Ginda, M. Data visualization literacy: Definitions, conceptual frameworks, exercises, and assessments. *Proc. Natl. Acad. Sci.* **116**, 1857–1864 (2019).